Machine learning analysis of extracellular vesicles for detection of triple negative breast cancer

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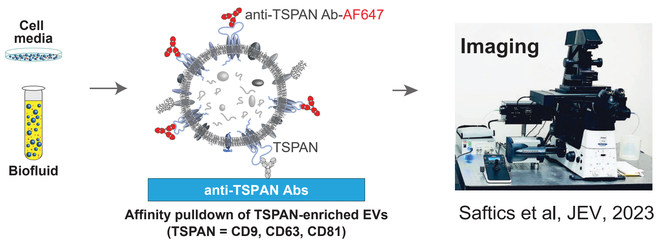
Abstract

Triple-negative breast cancers (TNBCs) are aggressive, often present at a more advanced stage, and frequently metastasize to distant organs. Accurate early diagnosis and longitudinal monitoring are critical for guiding treatment decisions. However, intratumoral heterogeneity and temporal changes in biomarker expression during disease progression and recurrence present major challenges. As therapies continue to evolve, a major clinical priority is to monitor dynamic changes in TNBC. Towards this, we are developing a liquid biopsy biomarker assay for early detection and monitoring of TNBC. From plasma of patients with TNBC (vs control), in a pilot study we evaluated small particles released from cells called extracellular vesicles (EVs). EVs are attractive biomarker sources because they carry molecular cargo reflective of their cells of origin and rapidly mirror changes in cellular state. However, TNBC-associated vesicles are rare in circulation. To address this, we employed a single EV imaging platform that can robustly isolate and quantify disease-relevant EV subpopulations from low plasma volumes. Specifically, we assessed EVs enriched in markers often associated with TNBC, including Trop2, PD-L1, and Complement 3. According to ensemble-based machine learning analysis of single EV imaging data, Complement 3-enriched EVs appeared to be promising markers for TNBC; they maintained a favorable balance between cancer detection and healthy sample recall. In contrast, incorporating PD-L1–enriched EVs reduced classification performance by over-predicting TNBC and diminishing healthy sample identification. Overall, our framework of EV-based liquid biopsy, which combines single-EV imaging with interpretable and trustworthy artificial intelligence, has the potential to detect rare TNBC-associated EV populations and support future early detection and non-invasive longitudinal monitoring of disease.

Single Extracellular VEsicle Nanoscopy (SEVEN)

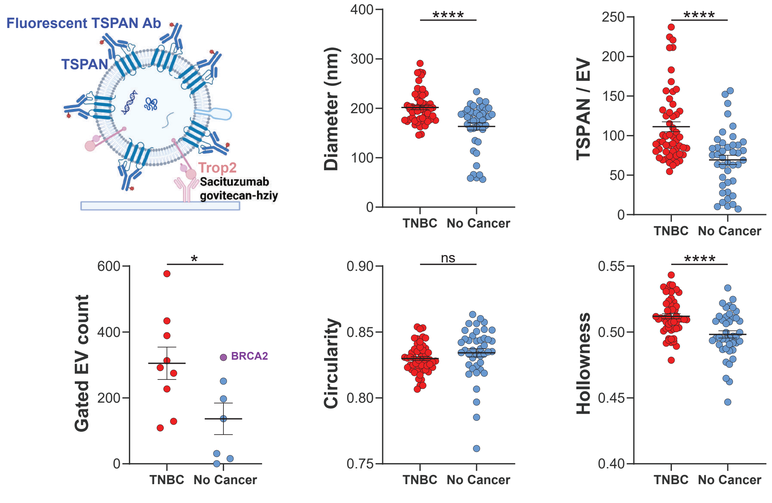


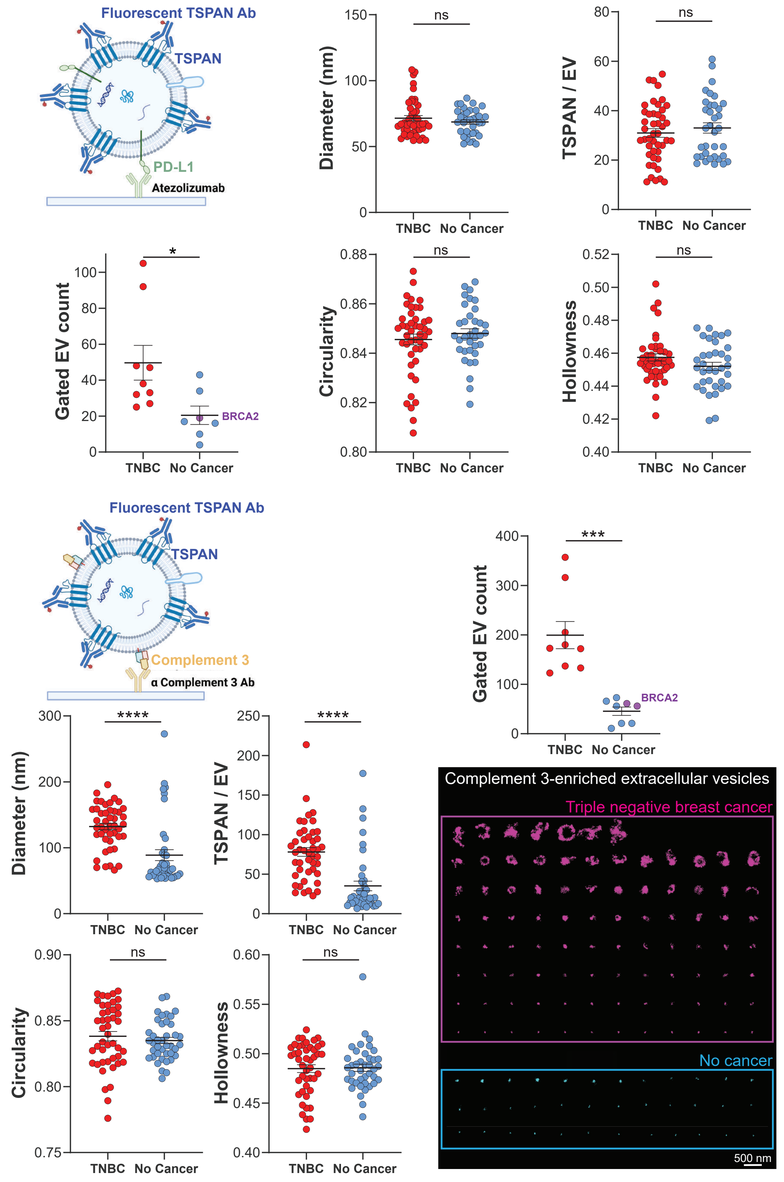
Affinity pulldown of TSPAN-enriched EVs (TSPAN = CD9, CD63, CD81)

Saftics et al, JEV, 2023

EVs are affinity isolated onto coverslips functionalized with primary antibodies (Ab) to enrich for specific populations and stained with the fluorescently labeled Abs to detect EVs and characterize their shape, size, and molecular content. In this example we used a mixture of anti-tetraspanin (anti-TSPAN) Abs to both isolate and detect EVs. Individual EVs are imaged using single molecule localization microscopy (SMLM) and data are rapidly analyzed using Voronoi Tesseleation algorithm.

Unique population of Trop2, PD-L1, and Complement 3 -enriched EVs in TNBC vs No Cancer





Trop2-enriched EVs from plasma of TNBC subjects (vs no cancer controls) were on average larger with more TSPAN molecules per EV and higher hollowness.

PD-L1 enriched EVs did not show significant difference in propretie between TNBC and healthy subjects.

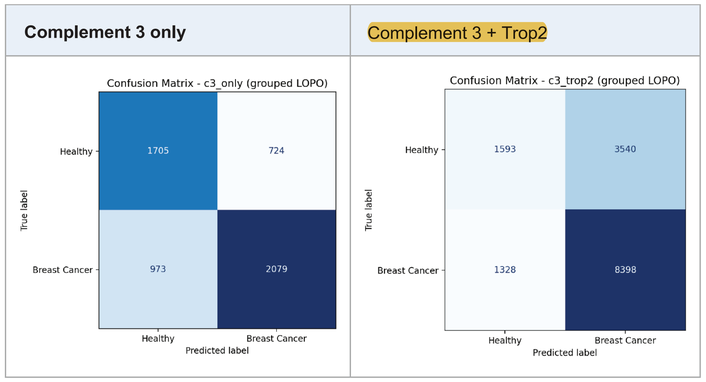
Complement 3-enriched EVs from palsma of TNBC patients (vs no cancer controls) were on average larger with more TSPAN molecules per EV; they often showed complex morphilogy.

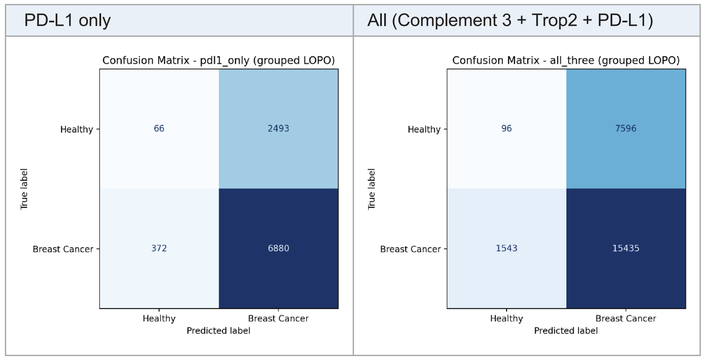
We detected significantly more gated EVs enriched in Trop2, PD-L1, and complement 3 in TNBC vs no conacer.

Machine learning identified Complement 3 as the best marker for TNBC vs. No Cancer

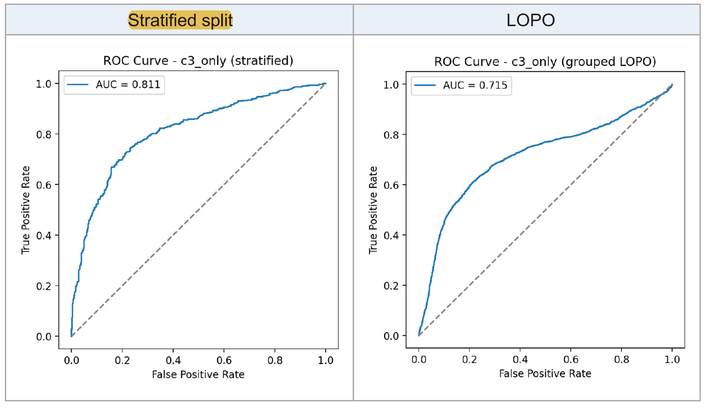
Leave One Patient Out (LOPO)

Marker	Accuracy	Healthy Recall	BC Recall	AUC
Complement 3 only	**0.690**	**0.702**	**0.681**	**0.715**
Trop2 + PD-L1	0.718	0.003	0.988	0.235
PD-L1 only	0.708	0.026	0.949	0.289
Trop2 only	0.701	0.003	0.984	0.270
Complement 3 + Trop2	0.672	0.310	0.863	0.518
All	0.630	0.012	0.909	0.264
Complement 3 + PD-L1	0.558	0.024	0.816	0.267



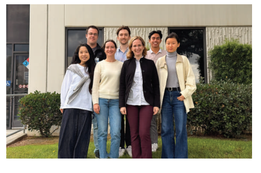


According to ensemble-based machine learning analysis of single EV imaging data, Complement 3-enriched EVs maintained a favorable balance between cancer detection and healthy sample recall. Addition of Trop2-enriched EVs reduced healthy sample recall, while addition of PD-L1-enriched EVs almost completely abolished healthy sample recall.



Conclusions and Future Directions

Our EV-based liquid biopsy framework could detect rare TNBC-associated EVs. We identified Complement 3-enriched EVs as a promising marker for TNBC, with a balanced detection of both cancer and control subjects. In the future, we will extend our analysis of Complement 3-enriched EVs to include 1) additional TNBC patients with varrying tumor size, 2) Luminal A and Luminal B breats cancer patients, 3) additional no cancer controls that will include BRCA1 and BRCA2 carriers.



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